Game Theory Week 1

Problem 1

$BR_1(L) = B$
$BR_1(R) = T$

$BR_2(T) = R$
$BR_2(B) = L$

$$NE = (B, T) or (R, L)$$

$$BR_1(L) = M, BR_1(C) = T, BR_1(R) = B$$

$$BR_2(T) = L, BR_2(M) = C, BR_2(B) = R$$

$$NE = (R, B)$$

Problem 3

The following are the utility functions of the Labour and Conservative parties.

$$u_L(x, y) = \frac{x}{x+y} - x, \quad u_c(x, y) = \frac{y}{x+y} - y$$

1. Show that Labour's best-reply function is given by $B_L(y) = \sqrt{y} - y$. What is the Conservative party's best-reply function?

We want to maximize the share of votes x as a function of y, so we take the partial derivative with respect to x and set it equal to zero.

$$\frac{\partial u_L}{\partial x} = \frac{1(x+y) - x(1)}{(x+y)^2} - 1 = \frac{y}{(x+y)^2} - 1$$

$$\frac{y}{(x+y)^2} - 1 = 0$$

$$y = (x+y)^2$$

$$x = \sqrt{y} - y$$

The second derivative of this

Therefore Labour's best response function $B_L(y) = \sqrt{y} - y$.

We can do the same for the Conservative parties best response function. This time I will differentiate with respect to y and solve for y as a function of x.

$$\frac{\partial u_C}{\partial y} = \frac{1(x+y) - y(1)}{(x+y)^2} - 1 = \frac{x}{(x+y)^2} - 1$$

$$x = (x+y)^2$$

$$y = \sqrt{x} - x$$

Therefore conservatives best reply function $B_C(x) = \sqrt{x} - x$.

2. The Nash equilibrium can be found at the point where $B_C(x^*) = y^*$ and $B_L(y^*) = x^*$.

$$x = \sqrt{y} - y$$

$$y = \sqrt{x} - x$$

By substitution

$$\sqrt{\sqrt{x} - x} - (\sqrt{x} - x) - x = 0$$

$$\sqrt{\sqrt{x} - x} - \sqrt{x} = 0$$

Using a graphing calculator I arrive at the solution

$$x^* = 0.25$$

Plugging back into the best-response function for the conservative party.

$$y^* = \sqrt{x} - x = \sqrt{0.25} - 0.25 = 0.25$$

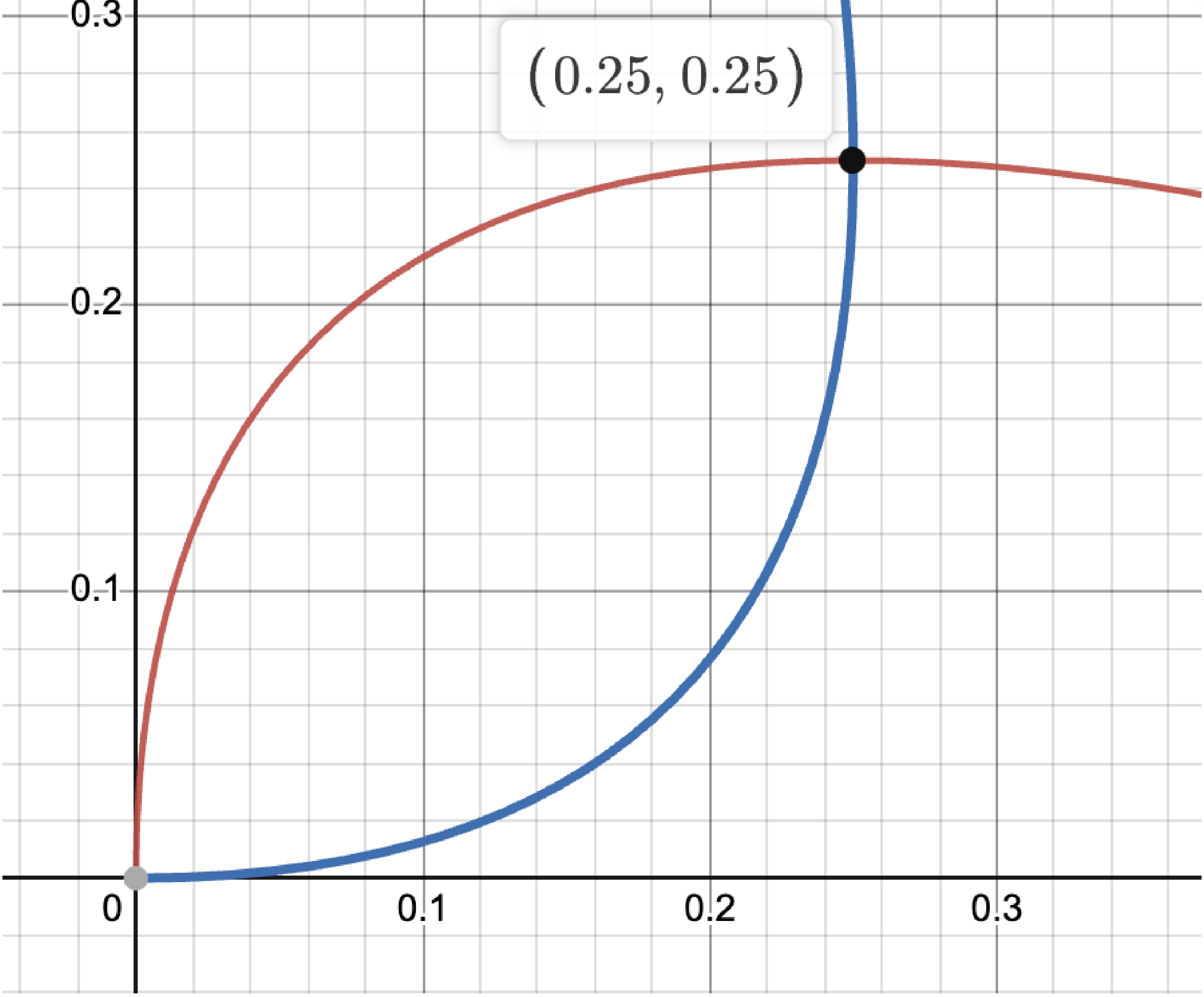
There is also a solution of $y^* = 0, x^* = 0$ but for this part we assume that $x > 0, y > 0$.

Given the Nash equilibrium of $(x^* = 0.25, y^* = 0.25)$ the vote shares for each party would be

$$v_L(y^*, x^*) = \frac{0.25}{0.25 + 0.25} = 0.5, v_C(y^*, x^*) = \frac{0.25}{0.25 + 0.25} = 0.5$$

Both parties would receive exactly 50% of the vote.

3. Plot the best-reply functions, and identify the equilibrium.



There are two equilibriums at $(x^* = 0, y^* = 0)$ and $(x^* = 0.25, y^* = 0.25)$.